Chapter 9: Light – Reflection and Refraction

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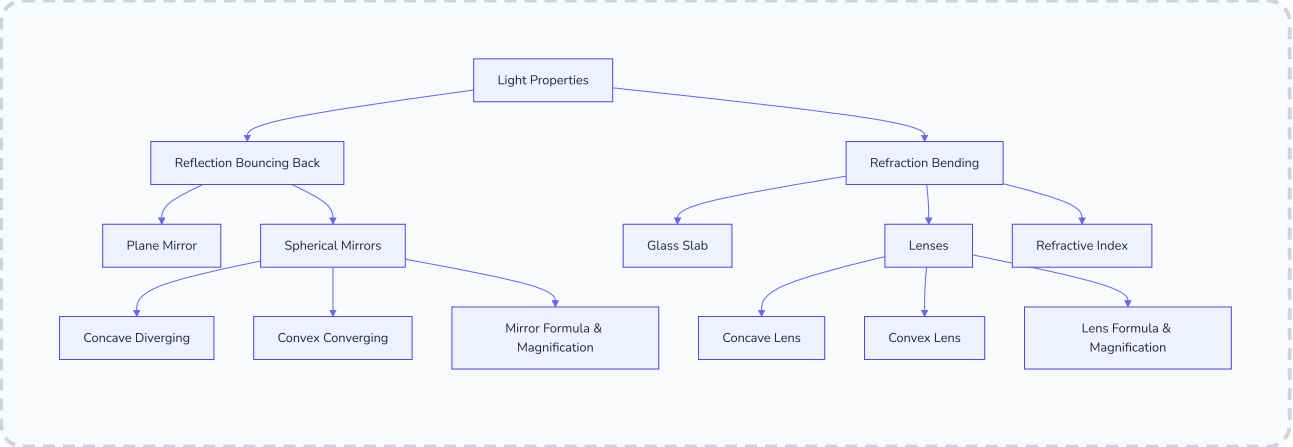
Light enables us to see things around us. It is a form of electromagnetic energy that travels in straight lines. In this chapter, we explore how light behaves when it hits a beautifully polished surface (Reflection) and when it crosses into a different medium like water or glass (Refraction).

Master Mind Map 🗺️



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1. Reflection of Light 🔮

Reflection is the bouncing back of light rays when they hit a highly polished surface like a mirror.

Laws of Reflection

1. The angle of incidence is EQUAL to the angle of reflection. ($\angle i = \angle r$)
2. The incident ray, the reflected ray, and the normal at the point of incidence, all lie in the **same plane**.

Image Formed by a Plane Mirror

- **Virtual and Erect** (Cannot be obtained on a screen, stands upright).
- **Same size** as the object.
- **Laterally inverted** (Left appears right and vice versa).
- Distance behind the mirror is equal to distance in front of it.

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2. Spherical Mirrors 🥄

Mirrors whose reflecting surfaces are curved inwards or outwards.

Feature	Concave Mirror	Convex Mirror
Shape	Reflecting surface curved INWARDS.	Reflecting surface curved OUTWARDS.
Nature	Converging Mirror (Brings parallel rays together to a Focus).	Diverging Mirror (Spreads parallel rays apart).
Focal Length	Real Focus (In front of mirror).	Virtual Focus (Behind the mirror).
Uses	Torches, vehicle headlights, shaving mirrors, dentist mirrors.	Rear-view mirrors in vehicles, security mirrors in shops.

QUICK TIP 💡

> **Why use Convex Mirrors in vehicles?** Because they always form an erect (upright) and diminished (smaller) image, providing the driver with a much wider field of view of the traffic behind!

Mirror Formula and Magnification

The mirror formula relates the object distance (u), image distance (v), and focal length (f).

$$ \frac{1}{v} + \frac{1}{u} = \frac{1}{f} $$

Magnification (m): The ratio of the height of the image (h') to the height of the object (h). $$ m = \frac{h'}{h} = -\frac{v}{u} $$

> ⚠ **COMMON MISTAKES** ❌
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> **Sign Convention Rules (Cartesian):**
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> - Object is always placed on the left (u is always negative).
> - Focal length of Concave Mirror/Lens is **negative**.
> - Focal length of Convex Mirror/Lens is **positive**.

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3. Refraction of Light 🌊

Refraction is the bending of light when it passes obliquely from one transparent medium to another. This happens due to the **change in the speed of light** in different media.

Rules of Bending

- **Rarer to Denser medium** (e.g., Air to Glass): Light slows down and bends **towards** the normal.
- **Denser to Rarer medium** (e.g., Water to Air): Light speeds up and bends **away** from the normal.

Laws of Refraction

1. The incident ray, the refracted ray, and the normal to the interface at the point of incidence, all lie in the same plane.
2. **Snell's Law of Refraction**: The ratio of sine of angle of incidence to the sine of angle of refraction is a constant. $$ \frac{\sin i}{\sin r} = \text{constant} $$

Refractive Index (n) 💎

It is the measure of how much light bends in a medium. It is the ratio of the speed of light in a vacuum/air (c) to the speed of light in the medium (v). $$ n_m = \frac{\text{Speed of light in vacuum } (c)}{\text{Speed of light in medium } (v)} $$

- The absolute refractive index of Diamond is 2.42 (highest), meaning light travels very slowly inside a diamond!

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4. Spherical Lenses 🔍

A transparent material bound by two surfaces, of which one or both are spherical, forms a lens.

Feature	Convex Lens	Concave Lens
Shape	Thicker in the middle, thinner at edges.	Thinner in the middle, thicker at edges.
Nature	Converging Lens.	Diverging Lens.
Image formed	Can be Real/Virtual depending on distance.	ALWAYS Virtual, Erect, and Diminished.

Lens Formula and Magnification

$$ \frac{1}{v} - \frac{1}{u} = \frac{1}{f} $$

Magnification (m): $$ m = \frac{h'}{h} = \frac{v}{u} $$

Power of a Lens (P) ⚡

The degree of convergence or divergence of light rays achieved by a lens. It is the reciprocal of its focal length (in meters). $$ P = \frac{1}{f(\text{in meters})} $$

- SI Unit of Power: **Diopter (D)**. (1 Diopter = 1 m^{-1})
- Power of Convex Lens = Positive ($+$).
- Power of Concave Lens = Negative ($-$).

Quick Revision Notes 📌

1. **Reflection** obeys exactly two simple laws across all surfaces: Angle i = Angle r, and all rays + normal sit on one plane.
2. **Concave Mirrors** can burn paper because they converge all sun rays to a single point (Focus).
3. **Lateral Shift**: The perpendicular distance between the original incident ray and the emergent ray coming out of a glass slab.
4. **Snell's Law**: $\frac{\sin i}{\sin r} = n$
5. **Power unit is Diopter (D)**. An optician prescribing a lens of $+2.0$ D means the person needs a Convex lens with a focal length of $+0.5$ m.

Chapter Summary 🏁

Light is a straight-traveling energy that either bounces (Reflection) or bends (Refraction). Spherical mirrors (Concave/Convex) reflect light to converge or diverge it, making them useful for torches or car rear-views. When light enters a different medium like glass or water, its speed changes, causing it to refract. We use this principle to craft Lenses (Convex/Concave) which powerfully manipulate light for helping our vision, microscopes, and cameras. Formulas and sign conventions mathematically prove exactly where every focused image sits!